

Wide Field-of-View Mixed Reality

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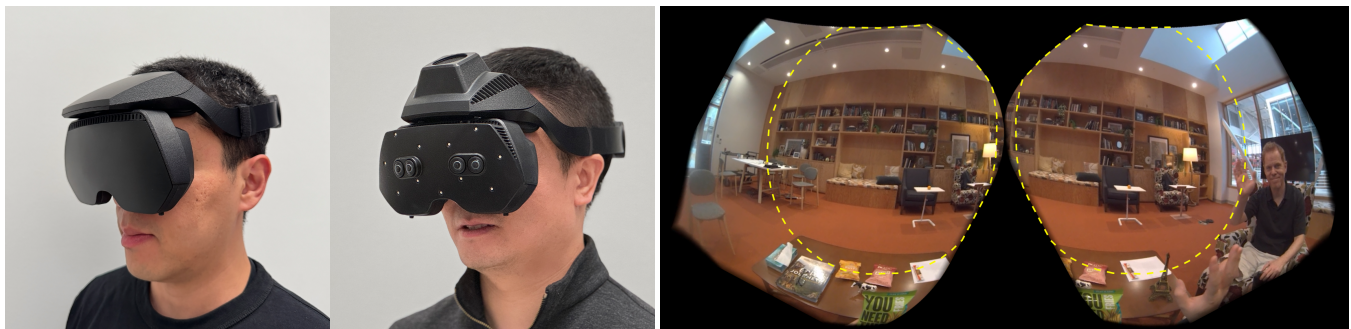


Figure 1: (Left) Photos of our wide field-of-view VR and MR prototypes are shown on the left and right, respectively. (Right) Screenshot showing MR passthrough images. The yellow dashed contours indicate the *Meta Quest 3* field of view.

Abstract

The human visual system has a horizontal field of view (FOV) of approximately 200 degrees. However, existing virtual and mixed reality headsets typically have horizontal FOVs around 110 degrees. While wider FOVs have been demonstrated in consumer devices, such immersive optics generally come at the cost of larger form factors, limiting physical comfort and social acceptance. We develop a pair of wide field-of-view headsets, each achieving a

horizontal FOV of 180 degrees with resolution and form factor comparable to current consumer devices. Our first prototype supports wide-FOV virtual reality using a custom optical design leveraging high-curvature reflective polarizers. Our second prototype further enables mixed reality by incorporating custom cameras supporting more than 80 megapixels at 60 frames per second. Together, our prototype headsets establish a new state-of-the-art in immersive virtual and mixed reality experiences, pointing to the user benefits of wider FOVs for entertainment and telepresence applications.

*Both authors contributed equally to this research.

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1 Introduction

The human visual system has a horizontal field of view (FOV) of approximately 200 to 220 degrees and a vertical FOV of 130 to 135 degrees [Howard and Rogers 1995]. Existing virtual and mixed reality head-mounted displays have significantly narrower FOVs than these perceptual limits. Current-generation products (e.g., *Meta Quest 3*, *Apple Vision Pro*, and *Valve Index*) have FOVs up to about 110 degrees (horizontal) and 100 degrees (vertical). Certain commercial HMDs that offer wider FOVs (e.g., Pimax Vision 8KX) feature FOVs up to about 160 by 100 degrees [Pimax 2025]. However, these HMDs incorporate optics and display components that generally result in significantly larger form factors, making them less suitable for prolonged wear and everyday use. Prototype wide-FOV VR displays have also been reported and exhibited; however, such prototypes either require unconventional components, such as curved display panels [Ratcliff et al. 2020] or require tiling multiple sets of displays and lenses, which often results in bulky form factors [Kishida et al. 2024]. To our knowledge, prior wide-FOV HMDs have not incorporated wide-FOV mixed reality (MR) passthrough, limiting applications to virtual reality (VR). In addition, the current commercial HMD with the widest MR FOV, i.e., *Varjo XR4*, provides 120-degree horizontal and 105-degree vertical FOV [Varjo 2025], significantly less than the FOV of our MR prototype.

In this installation we present a pair of novel wide-FOV HMDs, one supporting wide-FOV VR experiences — achieving the smallest-possible form factor with our optical design — and another supporting passthrough MR (see Fig. 1). Both headsets exceed state-of-the-art performance in FOV, form factor, and passthrough image quality. Specifically, our display system presents a 180-degree horizontal and 120-degree vertical FOV (see Fig. 2), with a resolution greater than 25 pixels/degree and with a form factor comparable to *Meta Quest 3*. In addition, our mixed reality prototype includes a high-bandwidth color camera system that achieves 33 pixels/degree.

2 VR Display System

Our wide-FOV virtual reality headset uses a pair of 2880×2880 LCD panels. The eyepieces are custom-designed pancake lenses that each include two refractive elements, with the primary novelty being the incorporation of a highly curved reflective polarizer (see Fig. 2). The optics and display modules for each eye are canted outward by 10 degrees, conforming to the user’s face and further assisting in expanding the field of view. Notably, these modules have a total optical track length of 27.3 mm, resulting in a form factor closely approximating current-generation products (e.g., *Meta Quest 3*).

3 MR Passthrough System

The wide-FOV and high resolution we targeted for our MR prototype created significant practical engineering challenges. To this end, we developed a custom passthrough camera system that contains four rolling-shutter color sensors, each with a resolution of 20 megapixels (5120×3840) and equipped with a fisheye lens. The cameras are positioned to roughly align with the separation between the user’s eyes, but are canted by 15 degrees, both outward and downward, to compensate for the asymmetry in the targeted display FOV. To enable high bandwidth streaming, we further developed a custom FPGA-based interface board that supports streaming up to 80

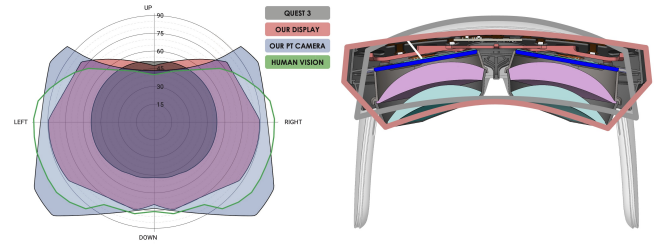


Figure 2: (Left) A comparison of the binocular fields of view of our prototype, *Meta Quest 3*, and human vision, assuming forward-directed gaze without eye rotation. The *Quest 3* FOV subtends approximately 46% of human vision in terms of solid angle. Our prototypes subtend approximately 90% of human vision. **(Right)** Comparison of the cross-sectional area of our base VR prototype (outlined in red), as compared to *Quest 3* (outlined in gray). Our demonstration emphasizes that wide-FOV optics can be designed to largely match the form factor of current-generation headsets, while significantly increasing the visual sense of immersion. To that end, we emphasize that our VR prototype weighs only 350 grams.

megapixels at 60 frames/second. This data is transmitted via an optical fiber from the FPGA to a desktop PC for post-processing before being sent to the displays (i.e., implementing the ISP algorithms, rolling shutter correction, and lens distortion correction).

4 Demo Experience

Our installation presents two stations that showcase our wide-FOV prototypes, as shown in Fig. 1 and the supplementary video: one area of the booth aims to demonstrate the best optimized form factor for virtual reality experiences, and the other, with video passthrough, aims to demonstrate immersive mixed reality experiences. Notably, both prototypes achieve FOVs that are close to the human limits when looking straight forward (with eye rotations further extending FOVs beyond the values cited above). During the demos, users can toggle between our full FOV and simulated FOVs of alternative systems to experience the visual and experiential differences, as shown in Figure 1.

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